



Worksheet- 10

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Subject Name: ADBMS

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1. Aim: Perform CRUD Operations using No-SQL Database: MongoDB

2. Objective:

- Create and manage databases and collections.
- Perform **CRUD operations** — Create, Read, Update, and Delete documents.
- Use **aggregation** for grouping and summarizing data.
- Understand how to modify, query, and manage collections efficiently in MongoDB.
- Gain practical experience with real-world data handling through command-line operations.

3. Code:

```
// 1. Show all databases
```

```
show dbs
```

```
// 2. Create or switch to your database
```

```
use db_ravi_prince
```

```
// 3. Check current database
```

```
db
```

```
// 4. Create a collection (optional)
```

```
db.createCollection("movies")
```

```
// 5. Insert Operations
```

```
db.movies.insertOne({
```

```
  title: "Dream Code",
```

```
  director: "Ravi",
```

```
  year: 2010,
```

```
  rating: 8.8,
```

```
  genre: "Sci-Fi",
```

```
  features: ["Mind-bending", "Action"],
```

```
  added_by: "Ravi"
```

```
})
```



```
db.movies.insertMany([
  {
    title: "Galaxy Journey",
    director: "Prince",
    year: 2014,
    rating: 8.6,
    genre: "Sci-Fi",
    features: ["Space", "Drama"],
    added_by: "Prince"
  },
  {
    title: "Ocean World",
    director: "Prince",
    year: 2009,
    rating: 7.8,
    genre: "Adventure",
    features: ["3D", "Action"],
    added_by: "Prince"
  },
  {
    title: "The Shadow Hero",
    director: "Ravi",
    year: 2008,
    rating: 9.0,
    genre: "Action",
    features: ["Hero", "Crime"],
    added_by: "Ravi"
  }
])
```

```
// 6. Show all collections
show collections
```

```
// 7. Read Operations
db.movies.find().pretty()
db.movies.findOne()
db.movies.find({}, { title: 1, year: 1, _id: 0 })
db.movies.find({ genre: "Action" })
```



```
db.movies.find({ "details.language": "English" })
```

```
// 8. Update Operations
```

```
db.movies.updateOne(  
  { title: "Ocean World" },  
  { $set: { rating: 8.1, genre: "Sci-Fi Adventure" } }  
)
```

```
db.movies.updateOne(  
  { title: "Dream Code" },  
  { $push: { features: "Thriller" } }  
)
```

```
db.movies.updateOne(  
  { title: "Dream Code" },  
  { $pull: { features: "Action" } }  
)
```

```
db.movies.updateMany(  
  { director: { $in: ["Ravi", "Prince"] } },  
  { $set: { language: "English" } }  
)
```

```
db.movies.updateMany({}, { $unset: { added_by: "" } })  
db.movies.updateMany({}, { $set: { quality: "Full HD" } })
```

```
db.movies.updateOne(  
  { title: "Time Reversal" },  
  { $set: { director: "Ravi", year: 2020, rating: 7.5 } },  
  { upsert: true }  
)
```

```
// 9. Delete Operations
```

```
db.movies.deleteOne({ title: "Ocean World" })  
db.movies.deleteMany({ director: "Ravi" })  
db.movies.deleteMany({})
```

```
// 10. Grouping (Aggregation)
```

```
db.movies.aggregate([
```

```
{ $group: { _id: "$director", total_movies: { $sum: 1 } } }  
])
```

```
db.movies.aggregate([  
  { $group: { _id: "$genre", total_movies: { $sum: 1 } } }  
])
```

```
// 11. Drop collection  
db.movies.drop()
```

```
// 12. Drop database  
db.dropDatabase()
```

Output:

```
admin          0.000GB  
college        0.000GB  
collegedb      0.000GB  
config         0.000GB  
db_company     0.000GB  
education_db   0.000GB  
library        0.000GB  
local          0.000GB  
switched to db db_ravi_prince  
db_ravi_prince  
{ "ok" : 1 }
```

```
{
  "acknowledged" : true,
  "insertedId" : ObjectId("690b89ab8090b47abf5502b5")
}
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("690b89ab8090b47abf5502b6"),
    ObjectId("690b89ab8090b47abf5502b7"),
    ObjectId("690b89ab8090b47abf5502b8")
  ]
}
movies
{
  "_id" : ObjectId("690b89ab8090b47abf5502b5"),
  "title" : "Dream Code",
  "director" : "Ravi",
  "year" : 2010,
  "rating" : 8.8,
  "genre" : "Sci-Fi",
  "features" : [
    "Mind-bending",
    "Action"
  ],
  "added_by" : "Ravi"
}
{
  "_id" : ObjectId("690b89ab8090b47abf5502b6"),
  "title" : "Galaxy Journey",
  "director" : "Prince",
  "year" : 2014,
  "rating" : 8.6,
  "genre" : "Sci-Fi",
  "features" : [
    "Space",
    "Drama"
  ],
  "added_by" : "Prince"
}
```

```
{
  "_id" : ObjectId("690b8ae81aec4a29ccce5507"),
  "title" : "Ocean World",
  "director" : "Prince",
  "year" : 2009,
  "rating" : 7.8,
  "genre" : "Adventure",
  "features" : [
    "3D",
    "Action"
  ],
  "added_by" : "Prince"
}
{
  "_id" : ObjectId("690b8ae81aec4a29ccce5508"),
  "title" : "The Shadow Hero",
  "director" : "Ravi",
  "year" : 2008,
  "rating" : 9,
  "genre" : "Action",
  "features" : [
    "Hero",
    "Crime"
  ],
  "added_by" : "Ravi"
}
{
  "_id" : ObjectId("690b8ae81aec4a29ccce5505"),
  "title" : "Dream Code",
  "director" : "Ravi",
  "year" : 2010,
  "rating" : 8.8,
  "genre" : "Sci-Fi",
  "features" : [
    "Mind-bending",
    "Action"
  ],
  "added_by" : "Ravi"
}
```

```
{ "title" : "Dream Code", "year" : 2010 }
{ "title" : "Galaxy Journey", "year" : 2014 }
{ "title" : "Ocean World", "year" : 2009 }
{ "title" : "The Shadow Hero", "year" : 2008 }
{ "_id" : ObjectId("690b8ae81aec4a29ccce5508"), "title" : "The Shadow Hero", "director" : "Ravi", "year" : 2008, "rating" : 9, "genre" : "Action", "features" : [ "Hero", "Crime" ], "added_by" : "Ravi" }
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
{ "acknowledged" : true, "matchedCount" : 4, "modifiedCount" : 4 }
{ "acknowledged" : true, "matchedCount" : 4, "modifiedCount" : 4 }
{ "acknowledged" : true, "matchedCount" : 4, "modifiedCount" : 4 }
{
  "acknowledged" : true,
  "matchedCount" : 0,
  "modifiedCount" : 0,
  "upsertedId" : ObjectId("690b8ae83b50a1c716af1df5")
}
{ "acknowledged" : true, "deletedCount" : 1 }
{ "acknowledged" : true, "deletedCount" : 3 }
{ "acknowledged" : true, "deletedCount" : 1 }
true
{ "ok" : 1 }
```



4. Learning Outcomes:

- Understand the structure and working of MongoDB as a NoSQL database.
- Create and manage databases and collections using MongoDB commands.
- Perform CRUD (Create, Read, Update, Delete) operations on documents.
- Apply aggregation operations for data grouping and analysis.
- Gain practical skills in handling and modifying data using MongoDB shell.